

SELU#

<https://pytorch.org/docs/stable/generated/torch.nn.SELU.html#torch.nn.SELU>

Description:

Applies the SELU function element-wise. with $\alpha=1.6732632423543772848170429916717$ and $\text{scale}=1.0507009873554804934193349852946$. When using `kaiming_normal` or `kaiming_normal_` for initialisation, `nonlinearity='linear'` should be used instead of `nonlinearity='selu'` in order to get Self-Normalizing Neural Networks. See `torch.nn.init.calculate_gain()` for more information. More details can be found in the paper [Self-Normalizing Neural Networks](#).

Function Signature

```
class torch.nn.SELU(inplace=False)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.SELU()
>>> input = torch.randn(2)
>>> output = m(input)
```

Notes & Warnings

WARNING

Warning When using `kaiming_normal` or `kaiming_normal_` for initialisation, `nonlinearity='linear'` should be used instead of `nonlinearity='selu'` in order to get Self-Normalizing Neural Networks. See `torch.nn.init.calculate_gain()` for more information.

Detailed Documentation

Documentation

Applies the SELU function element-wise.

with $\alpha=1.6732632423543772848170429916717$ $\alpha=1.6732632423543772848170429916717$
and $\text{scale}=1.0507009873554804934193349852946$ $\text{scale}=1.0507009873554804934193349852946$.

When using `kaiming_normal` or `kaiming_normal_` for initialisation, `nonlinearity='linear'` should be used instead of `nonlinearity='selu'` in order to get Self-Normalizing Neural Networks. See `torch.nn.init.calculate_gain()` for more information.

More details can be found in the paper [Self-Normalizing Neural Networks](#) .

`inplace` (bool, optional) – can optionally do the operation in-place. Default: False

Input: $(*)^{(*)}^{(*)}$, where $***$ means any number of dimensions.

Output: $(*)^{(*)}^{(*)}$, same shape as the input.

Return the extra representation of the module.

Runs the forward pass.